

GUI-Based Parking Management System

Developed using Python & Tkinter

Presented by: [Your Name]



Project Overview: Streamlining Parking

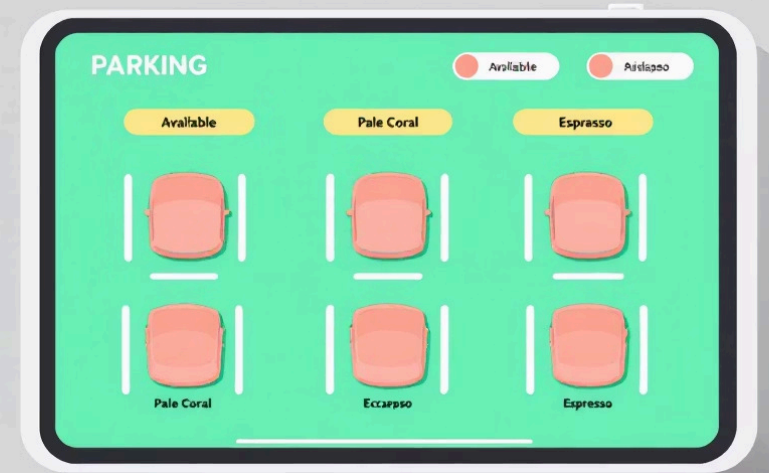
The Parking Problem

Finding available parking can be a daily hassle, leading to wasted time, frustration, and traffic congestion in urban areas and university campuses.



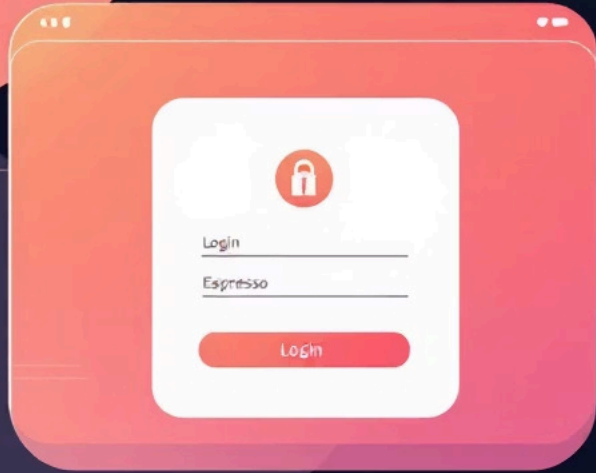
Our Automated Solution

This system automates parking management, providing a user-friendly interface to efficiently track and manage parking spots, enhancing overall convenience and reducing operational overhead.



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System Roles: Controlled Access



Administrator Role

Full system access, including management of parking spots and user data. Critical for system setup and oversight.

- Add/Edit/Delete spots
- View all system data

User Role

Limited access focused on vehicle management. Users can interact with their own car information and view assigned spots.

- Add/Edit/Delete cars
- View assigned spot

Access to features is dynamically controlled based on login credentials, ensuring secure and appropriate permissions for each role.

User Features: Seamless Car Management



Add Car

Automatically assigns an available parking spot upon vehicle entry.



Edit Car Info

Update vehicle details such as make, model, or license plate.



Delete Car

Removes vehicle data and frees up its assigned parking spot.



View Parked Cars

Browse a comprehensive list of all currently parked vehicles.

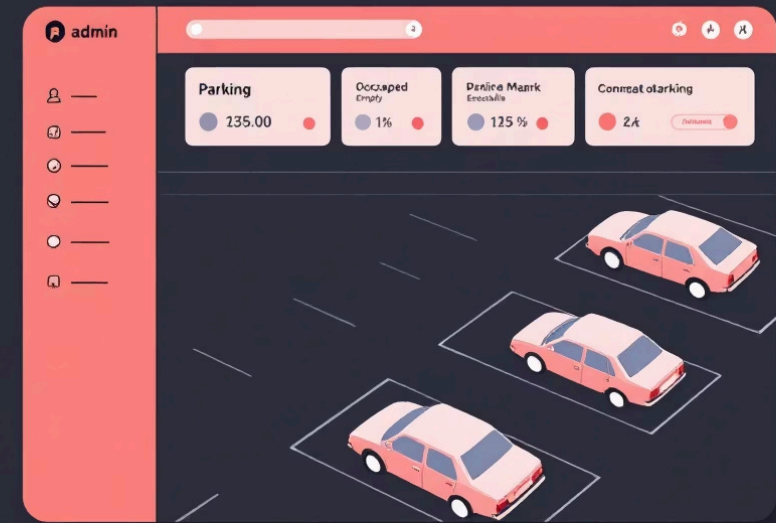


Search Cars

Quickly locate vehicles by name or license plate number.

Admin Features: Full Control & Oversight

- **Add Parking Spots** Dynamically expand parking capacity as needed.
- **Edit Spot Numbers** Modify existing spot identifiers for better organization.
- **Delete Empty Spots** Optimize layout by removing unused or redundant spots.
- **View Spot Occupancy** Real-time status of all parking spots, including details of parked vehicles.



The admin interface provides powerful tools for comprehensive parking facility management.

Advanced Features: Robust & Reliable

JSON Data Persistence

All system data is securely saved and loaded using JSON files, ensuring data integrity across sessions.

Input Validation

Robust checks prevent incorrect or malicious data entry, maintaining system accuracy.

Duplicate Prevention

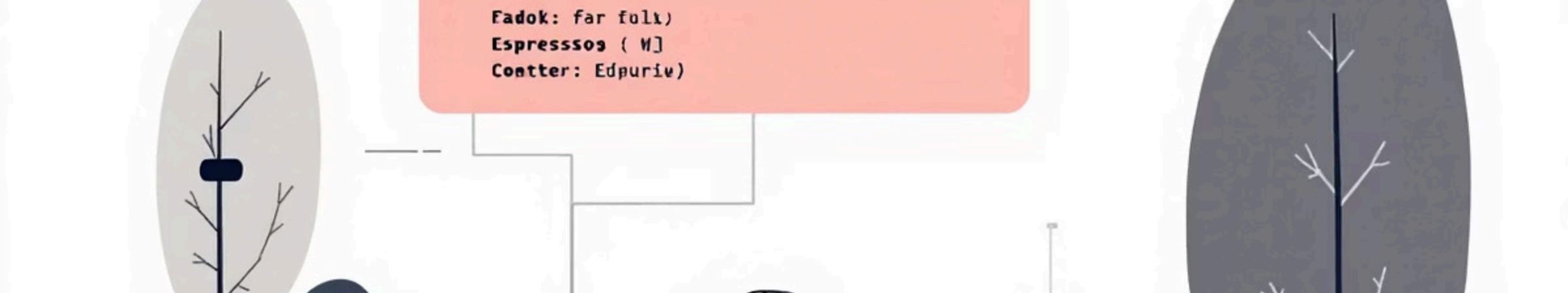
Ensures unique entries for vehicles and spots, avoiding conflicts.

Spot-to-Car Linking

Automatic and clear association between each parked car and its designated spot.

Parking Ticket Confirmation

Instant, clear messaging upon successful parking assignment for user reassurance.



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Data Storage: The Backbone of the System

parking_data.json

This central JSON file acts as the primary database for the system, storing all critical information. It includes:

- List of all defined parking spots
- Details of all currently parked cars
- Automatic updates to 'occupied' status as cars enter and exit.

Users List

Separate data manages user accounts and their associated roles (Admin/User), facilitating secure login and access control.

Technologies Utilized



Python

The core programming language for backend logic and system operations.



Tkinter

Used for developing the intuitive and responsive Graphical User Interface (GUI).



JSON File Handling

Enables efficient and persistent storage of all system data.

These technologies combine to create a robust, user-friendly, and maintainable parking management solution.

Conclusion: A Step Towards Smarter Parking

Project Demonstrations

- Practical application of GUI development.
- Effective data management and persistence.
- Implementation of role-based access control.

Skills Learned

- GUI design principles.
- Object-Oriented Programming (OOP) in Python.
- File I/O and data structuring.



Future Improvements & Enhancements



Mobile App Integration

Allow users to check spot availability and manage cars from their smartphones.



Online Payment Gateway

Integrate payment processing for parking fees.



Sensor-Based Spot Detection

Utilize physical sensors for real-time, highly accurate spot occupancy detection.



Advanced Reporting

Generate detailed reports on usage patterns, peak times, and revenue.

